

Predictive Forecasting of Care Load & Placement Demand

Abstract

The Unaccompanied Alien Children (UAC) Program operates in a highly dynamic environment where sudden changes in migration patterns, policy enforcement, and humanitarian crises can significantly affect care demand. This project develops predictive forecasting models to estimate future HHS care load and discharge demand using time-series and machine-learning techniques. Models including ARIMA, Random Forest, and Gradient Boosting were applied to forecast children in HHS care and identify potential capacity stress periods. The study demonstrates how predictive analytics can support proactive healthcare planning, resource allocation, and operational preparedness.

1. Introduction

The UAC Program provides temporary care, sheltering, medical support, and reunification services for children transferred from CBP custody into HHS care facilities. Due to the unpredictable nature of migration activity, the system experiences fluctuating intake volumes that can rapidly increase operational burden.

Traditional descriptive analytics explain historical trends but do not provide future operational intelligence. Therefore, predictive forecasting methods are necessary to help decision-makers anticipate future care demand and improve planning efficiency.

This project focuses on forecasting future HHS care load, discharge demand, and potential system imbalance using statistical and machine-learning models.

2. Objectives

Primary Objectives

- Forecast the number of children in HHS care
- Estimate future imbalance between intake and exits
- Predict short-term discharge demand

Secondary Objectives

- Provide early-warning insights for healthcare planners
- Compare statistical and machine-learning forecasting methods
- Improve proactive resource planning

3. Dataset Description

The dataset contains daily operational records from 2023–2025.

Column	Description
Date	Reporting date
Children apprehended and placed in CBP custody	Daily intake volume
Children in CBP custody	Active CBP care load
Children transferred out of CBP custody	Transfers into HHS
Children in HHS Care	Active HHS care load
Children discharged from HHS Care	Sponsor placements

4. Methodology

4.1 Time-Series Preparation

The Date column was converted into datetime format and arranged chronologically. Missing observations were handled using interpolation techniques to maintain continuity in daily records.

4.2 Feature Engineering

Several predictive features were created:

- Lag variables (t-1, t-7, t-14)
- Rolling averages and rolling variance
- Net pressure indicator:

Transfers – Discharges

- Calendar-based features such as weekday and month

4.3 Forecasting Models

The following forecasting approaches were implemented:

Baseline Models

- Naïve Forecast
- Moving Average Forecast

Statistical Models

- ARIMA
- Exponential Smoothing

Machine Learning Models

- Random Forest Regressor
- Gradient Boosting Regressor

4.4 Train-Test Strategy

A strict time-based split was used instead of random sampling to preserve temporal relationships within the data.

4.5 Evaluation Metrics

The models were evaluated using:

- MAE (Mean Absolute Error)
- RMSE (Root Mean Squared Error)
- MAPE (Mean Absolute Percentage Error)

These metrics measured forecast reliability and model performance.

5. Results and Findings

The forecasting models successfully identified future care-load trends and discharge demand patterns.

Machine-learning models such as Random Forest and Gradient Boosting demonstrated stronger performance compared to baseline forecasting methods, especially during periods of fluctuating intake.

ARIMA models effectively captured long-term trends and seasonality, while machine-learning models handled nonlinear variations more efficiently.

Forecasting results indicated periods where intake growth could potentially exceed discharge capacity, suggesting future operational stress and increased shelter burden.

6. Discussion

The study demonstrates the importance of predictive intelligence in healthcare and humanitarian operations management.

Forecasting future HHS care load allows decision-makers to anticipate overcrowding risk, allocate medical staff proactively, and improve shelter readiness before surge periods occur.

The net pressure indicator proved valuable for identifying future imbalance between incoming transfers and successful sponsor placements. Sustained positive pressure values may indicate potential backlog accumulation and longer shelter stays.

Comparisons between statistical and machine-learning approaches showed that no single model performs optimally under all conditions. Statistical models captured long-term trends effectively, while machine-learning models adapted better to sudden fluctuations.

Overall, predictive forecasting can significantly improve operational preparedness and support data-driven child welfare planning.

7. Recommendations

- Develop real-time forecasting dashboards
- Integrate early-warning capacity alerts
- Improve discharge planning workflows
- Increase staffing flexibility during forecasted surge periods
- Continuously retrain models using updated operational data

8. Conclusion

This project transformed the UAC dataset from a historical reporting system into a predictive intelligence framework. By applying time-series forecasting and machine-learning techniques, the study enabled proactive estimation of future care demand, discharge requirements, and operational stress periods.

The findings demonstrate that predictive analytics can improve healthcare planning, resource allocation, and humanitarian response efficiency within the UAC care system.

References

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